

PCA Model Evaluation and Comparison Report

Dataset: fraud_pca_95_variance.csv | Scope: trained fraud-detection models in this repository

Summary: Random Forest is the strongest trained model on the PCA feature set by PR-AUC (0.0313) and F1-score (0.0625). Overall scores remain low, which suggests PCA compression preserved broad variance but weakened minority-class fraud signal.

1. Evaluation Scope

This report compares every trained model available for the PCA dataset. The PCA dataset is treated as the feature-selected and PCA-analyzed representation of the fraud pipeline. It contains 11 principal-component features plus the binary fraud target. The same repository artifacts were used for the report; no model was retrained while compiling this document.

Item	Value
Dataset	PCA/fraud_pca_95_variance.csv
Feature representation	Wrapper-selected features transformed with PCA at 95% variance retention
Target	fraud (1 = suspected fraud, 0 = non-fraud)
Evaluated model families	MLP, RNN, Linear Regression, Decision Tree, Random Forest, XGBoost
Primary ranking metric	PR-AUC, because the fraud class is rare

2. Model Evaluation Results

Model	Family	Features	Threshold	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	Classical ML	11	0.5492	0.9160	0.0417	0.1244	0.0625	0.5633	0.0313
XGBoost	Classical ML	11	0.5781	0.9006	0.0364	0.1342	0.0573	0.5555	0.0288
Decision Tree	Classical ML	11	0.5719	0.7219	0.0268	0.3214	0.0494	0.5420	0.0263
MLP	Deep learning	11	0.5000	0.5490	0.0239	0.4778	0.0455	0.5232	0.0250
RNN	Deep learning	11	0.5000	0.5089	0.0244	0.5345	0.0467	0.5253	0.0243
Linear Regression	Classical ML	11	0.4994	0.5158	0.0239	0.5148	0.0456	0.5192	0.0238

Interpretation: accuracy is high for several tree models because the non-fraud class dominates. PR-AUC, recall, and F1 give a more useful view of fraud detection quality.

3. Confusion Matrix Summary

Model	TN	FP	FN	TP
Random Forest	32971	2321	711	101
XGBoost	32406	2886	703	109
Decision Tree	25801	9491	551	261
MLP	19434	15858	424	388
RNN	17941	17351	378	434
Linear Regression	18203	17089	394	418

4. Visual Comparison

The following chart compares the two most relevant aggregate metrics for rare-event fraud classification.

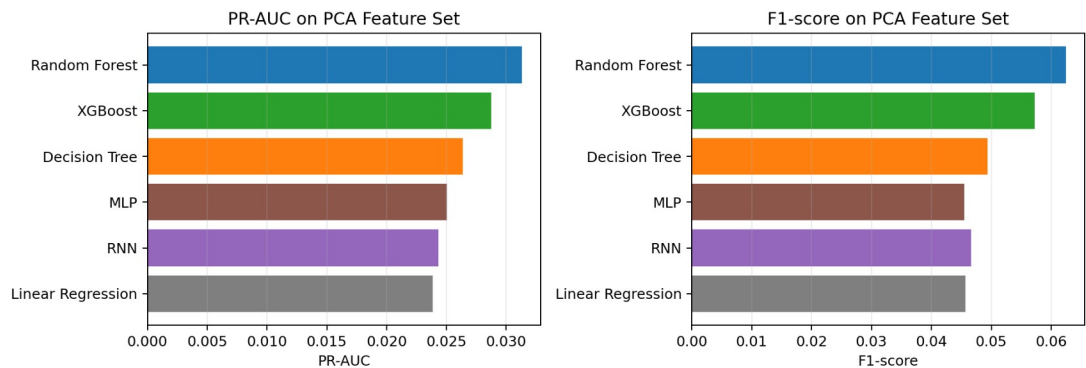


Figure 1. PR-AUC and F1-score comparison for all trained models on the PCA feature set.

The combined PR and ROC curves show how the model probability scores behave across thresholds.

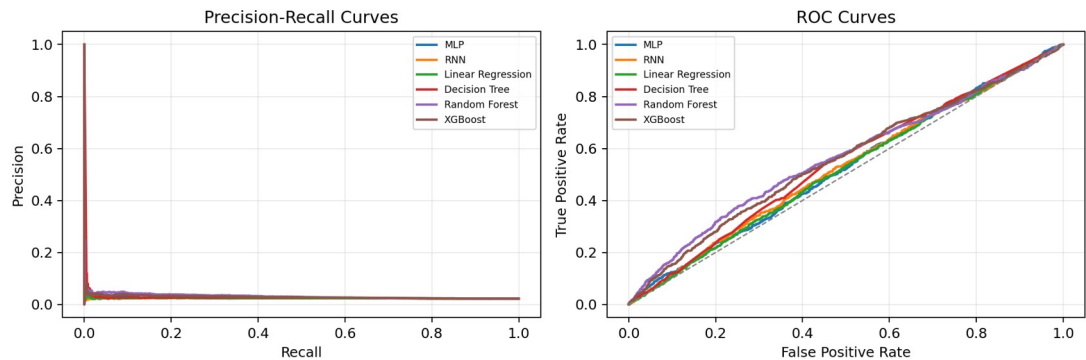


Figure 2. Precision-recall and ROC curves for PCA-trained models.

5. Findings

- Random Forest ranks first by both PR-AUC and F1-score on the PCA representation.
- XGBoost is the second strongest classical model by PR-AUC, but its recall and F1 remain modest.
- Decision Tree reaches the highest recall among the top three classical models, but it also creates many false positives.
- MLP and RNN perform close to the linear baseline on this PCA feature set, indicating that the PCA representation does not expose strong nonlinear fraud patterns to the neural models.
- Because fraud is rare, a model can show high accuracy while still missing most fraud cases. PR-AUC and F1 should be prioritized over accuracy in the final comparison.

6. Conclusion

For the PCA-analyzed fraud dataset, Random Forest is the recommended baseline among the currently trained models. However, all PCA-based scores are weak compared with the prepared numeric feature-space results already present in the repository, so the PCA representation should be treated as a compact experimental view rather than the best production feature set.